230701104

EXPERIMENT – 1

Installation and Configuration of Kali Linux in Oracle VirtualBox

Aim

To install and configure Kali Linux as a virtual machine using Oracle VirtualBox for performing network security and ethical hacking experiments.

Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux • System Requirements:
  - RAM: Minimum 4 GB (8 GB recommended) ○

  Disk Space: Minimum 30 GB ○ Processor:

  Intel/AMD with Virtualization support

Introduction:

Kali Linux is a Debian-based Linux distribution developed for penetration testing, digital forensics, and cybersecurity analysis.
Installing Kali Linux inside Oracle VirtualBox provides a safe and isolated environment to practice security experiments without affecting the host system.

VirtualBox is an open-source virtualization tool that allows running multiple operating systems simultaneously on a single physical machine.

Procedure

Step 1: Enable Virtualization

1. Restart the system and enter BIOS/UEFI settings.
2. Enable Intel VT-x / AMD-V.
3. Save changes and reboot the system.

Step 2: Install Oracle VirtualBox

1. Download Oracle VirtualBox from the official website.
2. Run the installer and follow default installation steps.
3. Complete the installation and restart the system if required.

Step 3: Download Kali Linux

1. Visit the official Kali Linux website.
2. Download the Kali Linux Installer ISO (64-bit).

Step 4: Create a New Virtual Machine

1. Open Oracle VirtualBox.
2. Click New.
3. Enter:
   - Name: Kali Linux
   - Type: Linux
   - Version: Debian (64-bit)
4. Click Next.

Step 5: Allocate Memory and CPU

- Memory Size: 2048 MB or higher
- Processor: 2 CPUs
- Click Next.

Step 6: Create Virtual Hard Disk

1. Select Create a virtual hard disk now.
2. Choose VDI (VirtualBox Disk Image).
3. Select Dynamically allocated.
4. Set disk size to 30 GB.
5. Click Create.

Step 7: Attach Kali Linux ISO 1. Select the created VM → Settings.

2. Go to Storage.
3. Under Controller IDE, select Empty.
4. Click Choose a disk file and select Kali Linux ISO.
5. Click OK.

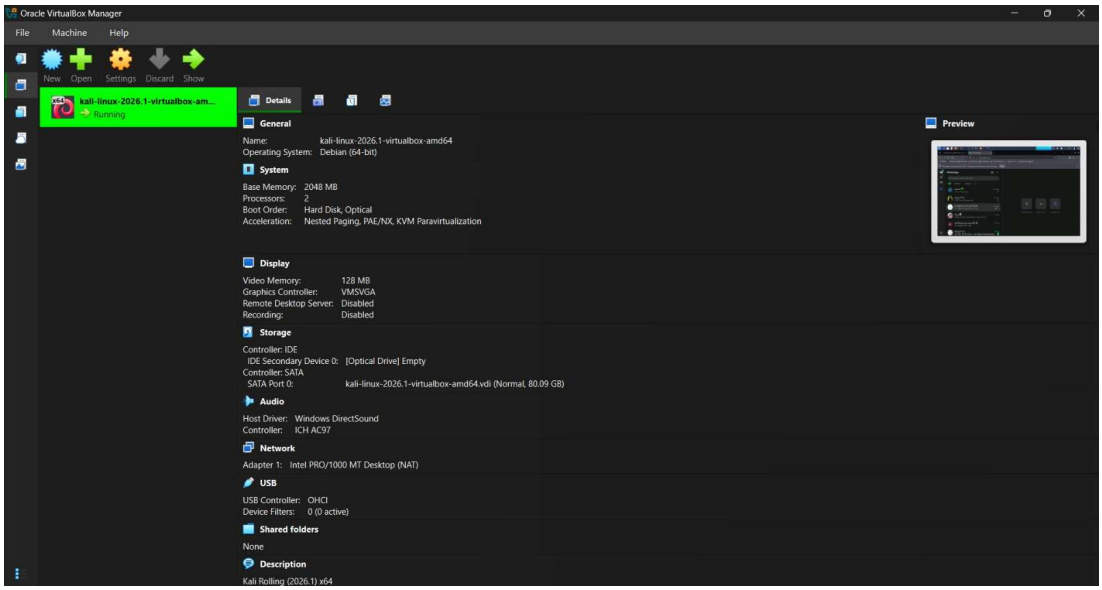
Step 8: Install Kali Linux 1. Start the virtual machine.

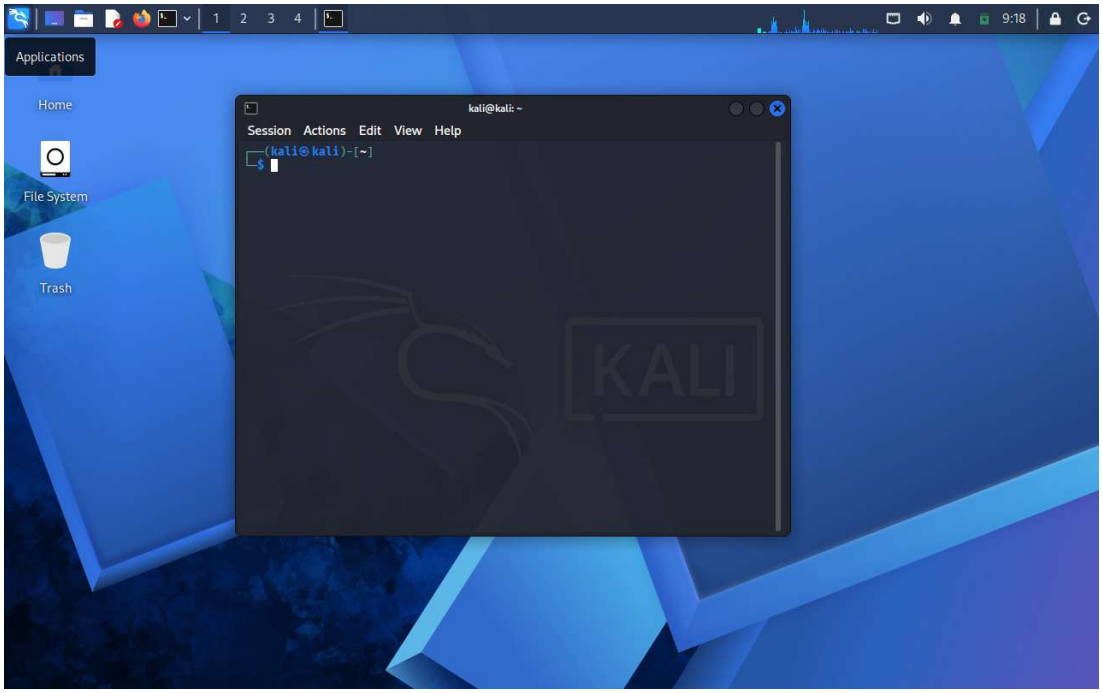
2. Select Graphical Install.
3. Choose:
   - Language
   - Location
   - Keyboard layout

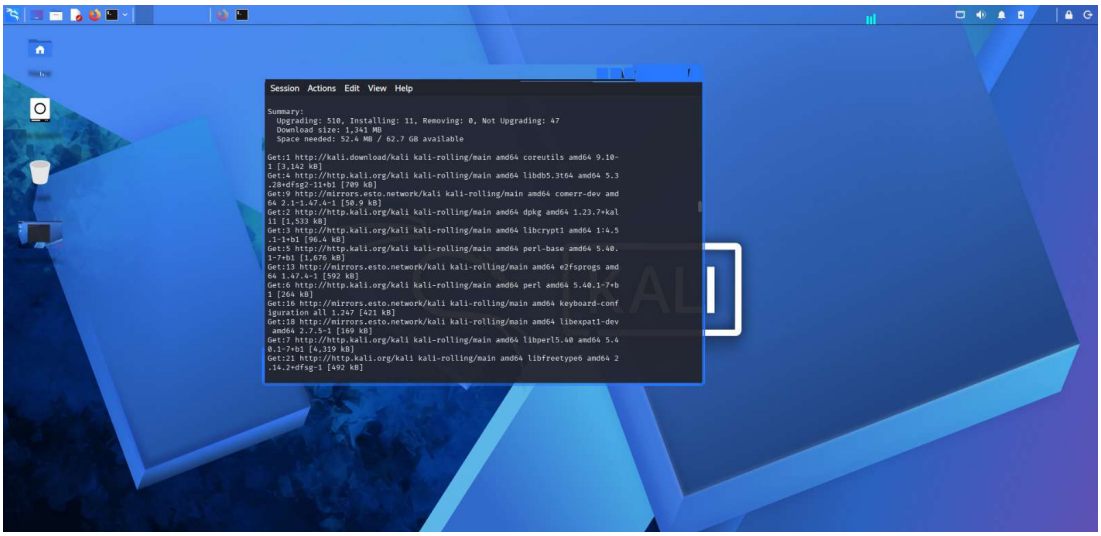
4. Configure network automatically.
5. Set:
   - Username
   - Password
6. Choose:
   - Guided – Use entire disk
   - All files in one partition

7. Confirm disk changes and continue installation.
8. After completion, reboot the system.

Output


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
kali-linux-2026.1-virtualbox-am...
Running
Details
General
Name: kali-linux-2026.1-virtualbox-amd64
Operating System: Debian (64-bit)
System
Base Memory: 2048 MB
Processors: 2
Boot Order: Hard Disk, Optical
Acceleration: Nested Paging, PAE/NX, KVM Paravirtualization
Preview
Display
Video Memory: 128 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: kali-linux-2026.1-virtualbox-amd64.vdi (Normal, 80.09 GB)
Audio
Host Driver: Windows DirectSound
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
Kali Rolling (2026.1) x64


1
2
3
4
9:18
Applications
Home
File System
Trash
kali@kali: ~
Session
Actions
Edit
View
Help
┌──(kali㉿kali)-[~]
└─$
KALI



Result

Kali Linux was successfully installed and configured in Oracle VirtualBox. The virtual machine is ready for performing network security and ethical hacking experiments.